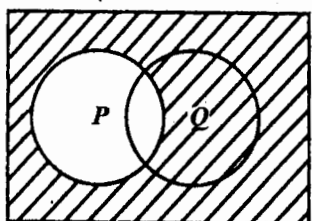


1. The exact value of $6 \div (0.0003)$ is
- (A) 200
(B) 2 000
(C) 20 000
(D) 200 000
2. 0.0346 written in standard form is
- (A) 3.46×10^2
(B) 3.46×10
(C) 3.46×10^{-1}
(D) 3.46×10^{-2}
3. Express 0.12 as a fraction in its lowest terms.
- (A) $\frac{1}{9}$
(B) $\frac{3}{5}$
(C) $\frac{1}{8}$
(D) $\frac{3}{25}$
4. $\sqrt{17^2 - 15^2}$
- (A) 1
(B) 2
(C) 8
(D) 16
5. John, Peter and Mary shared a sum of money in the ratio 2 : 4 : 9. John and Peter together received \$360. How much money in all was shared?
- (A) \$480
(B) \$540
(C) \$600
(D) \$900
6. Express $4\frac{3}{8}$ as a decimal correct to 3 significant figures
- (A) 4.30
(B) 4.37
(C) 4.38
(D) 4.40
7. What is the Highest Common Factor of the set of numbers {54, 72, 90}?
- (A) 9
(B) 18
(C) 90
(D) 1080
8. $(-\frac{1}{2})^3$ is the same as
- (A) $-\frac{1}{8}$
(B) $-\frac{1}{6}$
(C) $\frac{1}{8}$
(D) $\frac{1}{6}$
9. What is the value of the digit 2 in the number 48.621?
- (A) $\frac{2}{100}$
(B) $\frac{2}{10}$
(C) 2
(D) 200

10. By the distributive law $49 \times 17 + 49 \times 3 =$

- (A) $52 + 66$
- (B) 52×66
- (C) $49 + 20$
- (D) 49×20

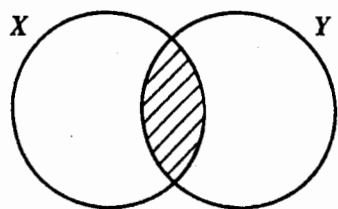
Item 11 refers to the Venn diagram below.



11. In the Venn diagram above, the shaded area represents

- (A) P'
- (B) $(P \cup Q)'$
- (C) $P' \cup Q$
- (D) $P' \cap Q$

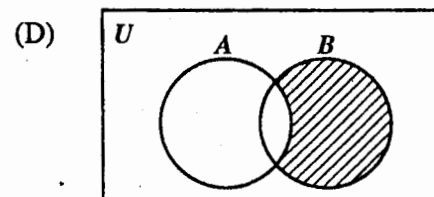
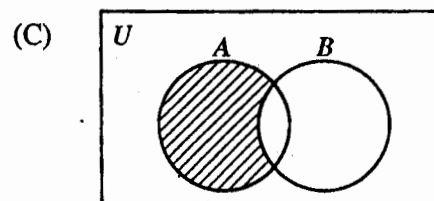
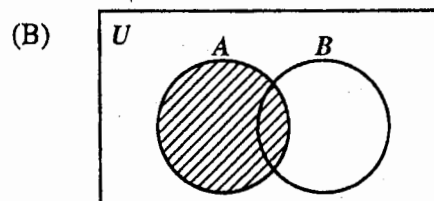
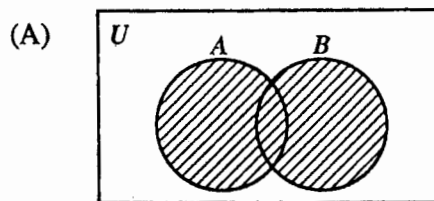
Item 12 refers to the Venn diagram below.



12. In the figure above, X represents the set of multiples of four. Y represents the set of multiples of five. The shaded region represents the set of all multiples of

- (A) 8
- (B) 9
- (C) 10
- (D) 20

13. In which one of the following Venn diagrams is the region $A \cap B'$ shaded?



14. $A = \{\text{Factors of } 30\}$
 $B = \{\text{Prime numbers less than } 10\}$
 $C = \{\text{Even whole numbers less than } 10\}$

Then $n(A \cap B \cap C)$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 9

15. If p sweets cost q cents, then the cost of one sweet is

- (A) $\frac{p}{q}$ cents
 (B) pq cents
 (C) $\frac{q}{p}$ cents
 (D) $(q - p)$ cents

16. A dress which costs \$180 is being sold at a discount of 10%. The amount of the discount is

- (A) \$ 1.80
 (B) \$ 10.00
 (C) \$ 18.00
 (D) \$170.00

17. An article bought for \$125 was sold for \$175. The percentage profit was

- (A) 28.6
 (B) 40
 (C) 50
 (D) 71.4

Item 18 refers to the table below.

Mark	0	1	2	3	4	5	6	7	8	9
No. of students	5	2	3	4	6	8	8	4	9	1

The table above shows the marks obtained by 50 students in a test. What is the probability that a student chosen at random would have obtained a score less than 5?

- (A) $\frac{2}{5}$
 (B) $\frac{3}{5}$
 (C) $\frac{11}{25}$
 (D) $\frac{14}{25}$

Item 19 refers to the table below.

House Insurance	25¢ per \$100
Contents Insurance	50¢ per \$100

19. The above table shows the rates charged by an insurance company. How much will a person pay for his insurance, if his house is valued at \$50 000, and the contents at \$10 000?

- (A) \$500
 (B) \$450
 (C) \$225
 (D) \$175

20. Tom bought a pen for \$60 and sold it to gain 20% on his cost price. How much money did he gain?

(A) \$12
(B) \$40
(C) \$72
(D) \$80

21. Mary invested \$200 for 3 years at 5% per annum. John invested \$300 at the same rate. If they both received the same amount of money in interest, for how many years did John invest his money?

(A) $1\frac{1}{2}$
(B) 2
(C) 3
(D) 10

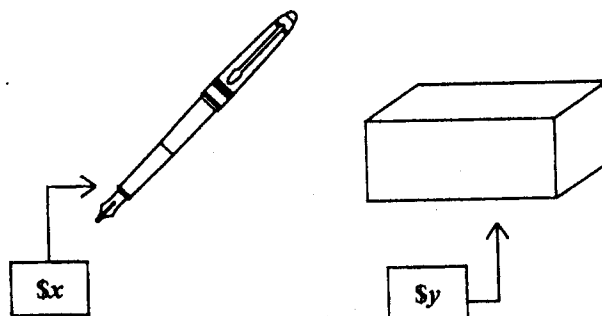
22. A woman buys a pair of shoes at a sale. She pays \$60, saving \$15 on the regular price. The percentage discount on the shoes is

(A) 20
(B) 25
(C) $33\frac{1}{3}$
(D) 80

23. $-2(x-4)$

(A) $-2x-8$
(B) $-2x-4$
(C) $-2x+4$
(D) $-2x+8$

Item 24 refers to the information below.



24. The total cost of 3 pens and 2 boxes is

(A) $6xy$
(B) $5(x+y)$
(C) $3x+2y$
(D) $2x+3y$

25. $3a(a+2b) - b(2a-3b) =$

(A) $3a^2 - ab + 3b^2$
(B) $3a^2 + 4ab + 3b^2$
(C) $3a^2 + 4ab - 3b^2$
(D) $3a^2 + 8ab - 3b^2$

26. Given that $a * b = 2a - 3b$, then $2 * (-3) =$

(A) 13
(B) 3
(C) -5
(D) -7

27. $2(a^2b)^3 =$

(A) $2a^3b^3$
(B) $2a^6b^3$
(C) $6a^2b$
(D) $8a^6b^3$

28. Given that $2x+3 \geq 9$, the range of values of x is

(A) $x > 3$
(B) $x \geq 3$
(C) $x > 6$
(D) $x \geq 6$

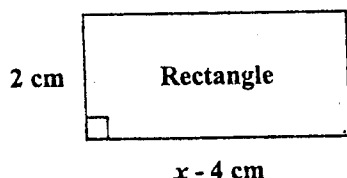
29. Given that $3(x - 1) - 2(x - 1) = 7$, the value of x is

(A) 6
(B) 7
(C) 8
(D) 9

30. John has x marbles and Max has twice as many. Max gives John 5 of his marbles. How many marbles does Max now have?

(A) $x + 5$
(B) $x - 5$
(C) $2x + 5$
(D) $2x - 5$

Item 31 refers to the following diagram.



31. The area of the rectangle, in cm^2 , is x^2 . The equation that may be used to find the value of x is

(A) $x^2 = 2(x - 4)$
(B) $x^2 = (x - 2)(x - 4)$
(C) $x^2 = 2(x - 4)(x - 2)$
(D) $x^2 = (x - 4)(x + 2)$

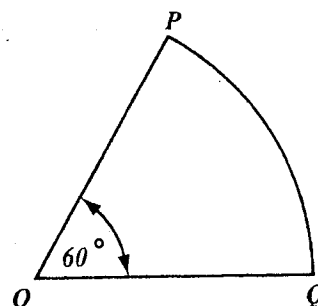
32. The mass, in kg, of a bag of rice is given as 10.6 correct to 1 decimal place. The range of values in which the actual mass lies is

(A) $10.5 \leq m \leq 10.7$
(B) $10.55 \leq m < 10.65$
(C) $10.59 \leq m \leq 10.69$
(D) $10.55 \leq m \leq 10.65$

33. The volume of a cube whose edge is 6 cm long is

(A) 18 cm^3
(B) 36 cm^3
(C) 72 cm^3
(D) 216 cm^3

Item 34 refers to the figure below.



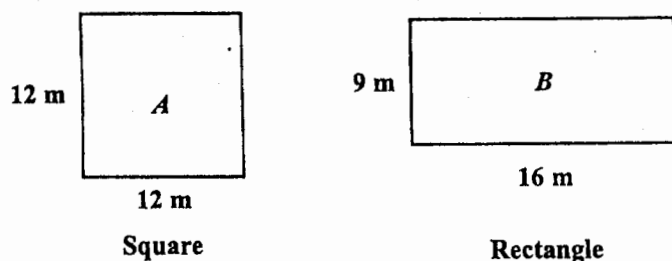
34. The figure above, not drawn to scale, shows a sector of a circle centre O . The length of the minor arc PQ is 8 cm. What is the length of the circumference of the entire circle?

(A) 16 cm
(B) 24 cm
(C) 48 cm
(D) 64 cm

35. The distance around the edge of a circular pond is 88 m. The radius, in metres is

(A) 88π
(B) 176π
(C) $\frac{88}{\pi}$
(D) $\frac{88}{2\pi}$

Item 36 refers to the diagrams below.



36. Which of the following statements is true about the perimeters of the figures A and B?

(A) Perimeter of A = Perimeter of B
 (B) Perimeter of A > Perimeter of B
 (C) Perimeter of A $\geq$ Perimeter of B
 (D) Perimeter of A < Perimeter of B

37. The lengths of the sides of a triangle are x , $2x$ and $2x$ centimetres. If the perimeter is 20 centimetres, what is the value of x ?

(A) 4
 (B) 5
 (C) 8
 (D) 10

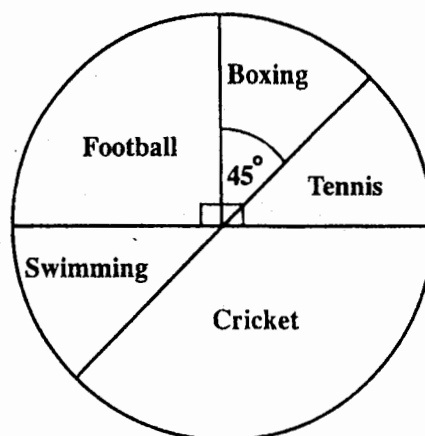
38. The area of a rectangle is 53.6 cm^2 . If the length is multiplied by four and the width is halved, the area would then be

(A) 26.8 cm^2
 (B) 53.6 cm^2
 (C) 107.2 cm^2
 (D) 214.4 cm^2

39. Each exterior angle of a regular polygon measures 45° . How many sides has the polygon?

(A) 10
 (B) 8
 (C) 6
 (D) 4

Items 40 - 42 refer to the diagram below which shows the sport chosen by 160 boys who participated in a games evening at their school.



40. The number of boys who chose football is

(A) 40
 (B) 90
 (C) 110
 (D) 150

41. The probability that a boy chosen at random participated in boxing is

(A) $\frac{1}{8}$
 (B) $\frac{1}{4}$
 (C) $\frac{1}{2}$
 (D) $\frac{7}{8}$

42. How many boys participated in cricket?

(A) 54
 (B) 60
 (C) 110
 (D) 120

43. Which one of the following is NOT a statistical diagram?

- (A) Bar graph
- (B) Pie chart
- (C) Frequency polygon
- (D) Modal class

44. A bag contains 2 red, 4 yellow and 6 blue balls. The probability of drawing a blue ball from the bag at random is

- (A) $\frac{1}{6}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{6}{11}$

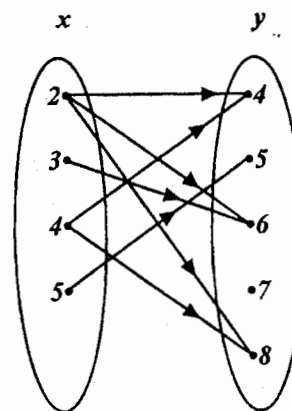
Item 45 refers to the table below which shows the frequency of scores obtained by students in a test.

Scores	2	3	5	6	8	11
Students	8	4	6	3	12	2

45. The modal score is

- (A) 8
- (B) 9
- (C) 10
- (D) 12

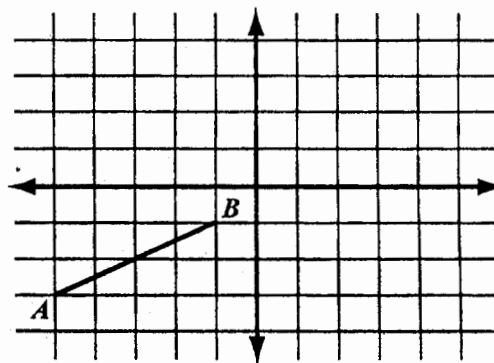
Item 46 refers to the arrow diagram below.



46. The arrow diagram above describes the relation

- (A) x is greater than y
- (B) x is a multiple of y
- (C) x is divisible by y
- (D) x is a factor of y

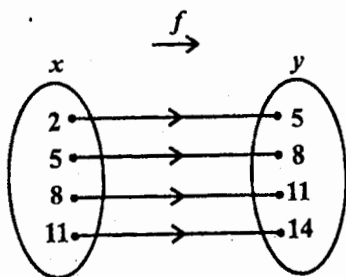
Item 47 refers to the figure below.



47. The gradient of AB in the figure above is

- (A) 2
- (B) $\frac{1}{2}$
- (C) $-\frac{1}{2}$
- (D) -2

48. The arrow diagram below shows a function.



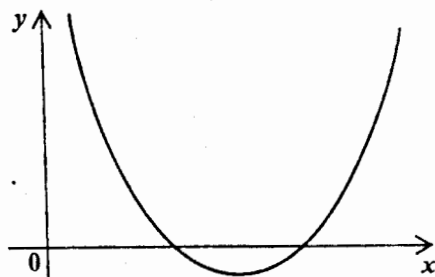
Which of the following BEST describes the function?

- (A) $f(x) = x + 3$
 (B) $f(x) = y + 3$
 (C) $x = y + 3$
 (D) $y = x$

49. The range of $f : x \rightarrow x^3$ for the domain $\{-2, -1, 0, 1, 2\}$ is

- (A) $\{0, 1, 8\}$
 (B) $\{-2, -1, 0, 1, 2\}$
 (C) $\{-6, -3, 0, 3, 6\}$
 (D) $\{-8, -1, 0, 1, 8\}$

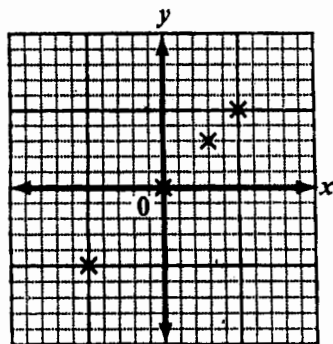
Item 50 refers to the following graph.



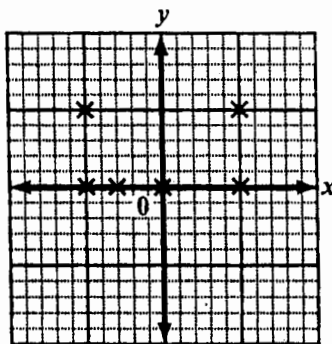
50. If a , b and c are constants, the equation of the graph could be

- (A) $y = ax^2 + c$
 (B) $y = c - ax^2$
 (C) $y = ax^2 + bx + c$
 (D) $y = c + bx - ax^2$

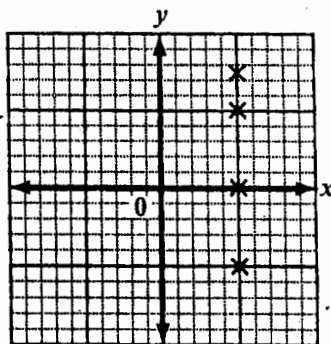
51. Which of the following represents the graph of a function?



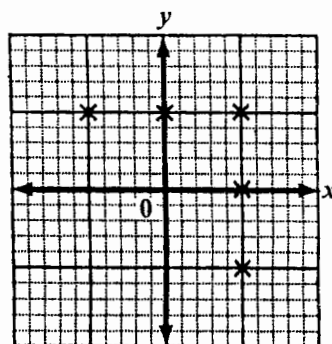
I



II



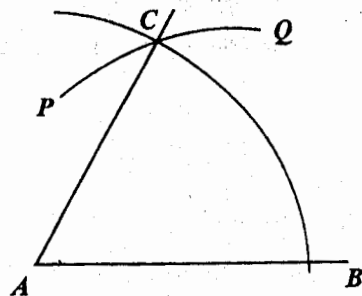
III



IV

- (A) I
 (B) II
 (C) III
 (D) IV

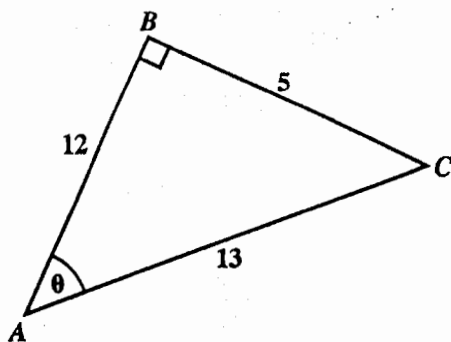
Item 52 refers to the diagram below of a construction. With centre A , an arc BC is drawn. With centre B , and the same radius, the arc PCQ is drawn.



52. What is the measure of $\angle BAC$?

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 75°

Item 53 refers to the right-angled triangle below.



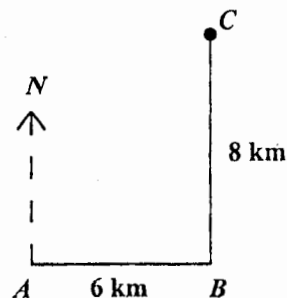
53. In the right-angled triangle above, $\tan \theta$ is

- (A) $\frac{5}{13}$
- (B) $\frac{5}{12}$
- (C) $\frac{12}{5}$
- (D) $\frac{13}{5}$

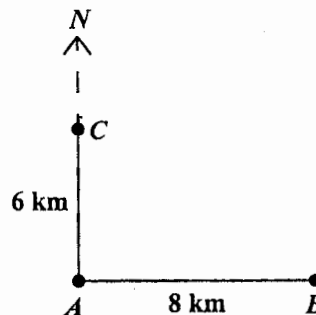
54.

A ship sails 8 km due east from A to B . It then sails 6 km due north to C . Which diagram below BEST represents the path of the ship?

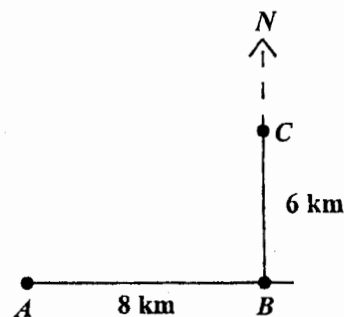
(A)



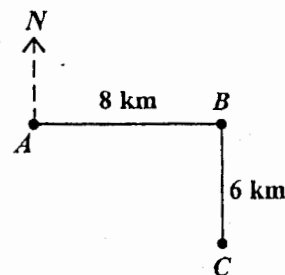
(B)



(C)



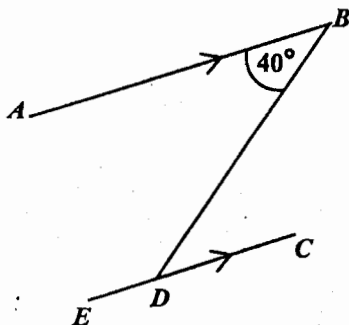
(D)



55. The image of a point $P(-2, 3)$ under a translation $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is

- (A) $(-6, 12)$
- (B) $(-5, -1)$
- (C) $(5, 1)$
- (D) $(1, 7)$

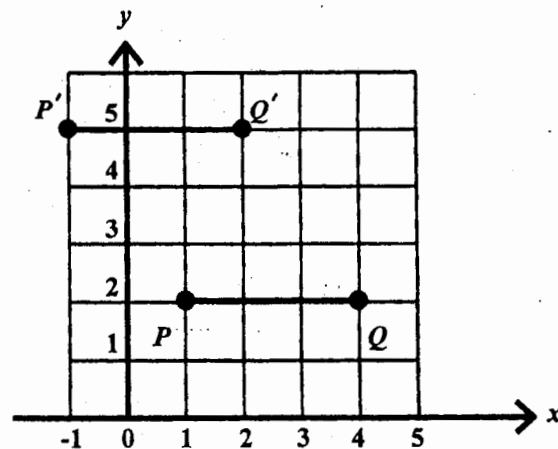
Item 56 refers to the diagram below.



56. In the diagram above, AB is parallel to EC . Calculate $\angle BDE$.

- (A) 40°
- (B) 50°
- (C) 140°
- (D) 180°

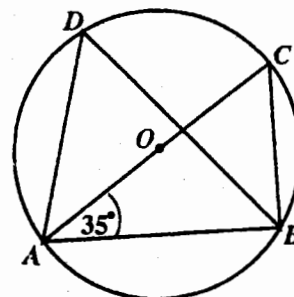
Item 57 refers to the diagram below.



57. In the diagram above, the vector that translates PQ to $P'Q'$ may be described as

- (A) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$
- (B) $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$
- (C) $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$
- (D) $\begin{pmatrix} -2 \\ -3 \end{pmatrix}$

Item 58 refers to the diagram below.



58. In the diagram above, $\angle CAB = 35^\circ$ and AC is the diameter of the circle. Angle ADB is

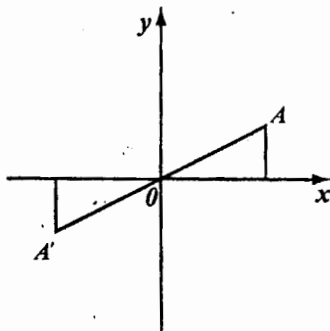
- (A) 55°
- (B) 45°
- (C) 35°
- (D) 65°

59. A plane is heading in a direction of 045° and changes course in a clockwise direction to 135° . The angle through which the plane turns is

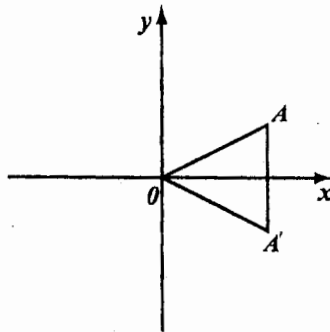
- (A) 45°
- (B) 90°
- (C) 135°
- (D) 270°

60. In each of the diagrams shown below, A' is the image of A . Which of the diagrams shows a reflection in the x axis?

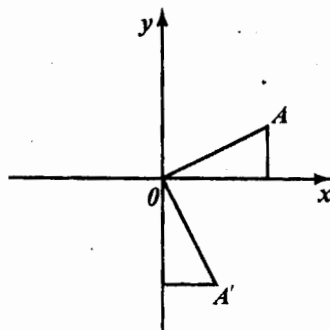
(A)



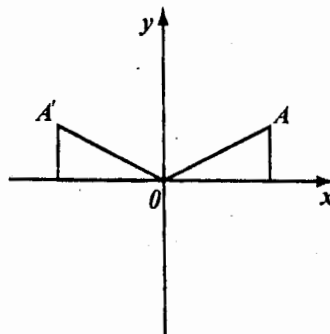
(B)



(C)



(D)



IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

